GOPIKA P S

Thiruvananthapuram, Kerala, India • +91 9446542070 • gopikaps011@gmail.com
Github: gop011 • LinkedIn: gopika-p-s-b96037298

Professional Summary

Electronics and Communication Engineering student with hands-on experience in analog/CMOS IC design, FPGA development, and embedded systems. Proficient in circuit simulation, custom design flows, and PDK-based analog design (SKY130), with working knowledge of microcontrollers (STM32, Arduino) and hardware description languages (Verilog). Seeking to apply strong analog design fundamentals and clear technical communication to an Applications Engineering role.

Education

B.Tech, Electronics & Communication Engineering Expected 2027
Government Engineering College, Sreekrishnapuram, Palakkad – APJ Abdul Kalam Technological University, Kerala
CGPA: 9.11 (up to semester 6)

Technical Skills

- **Analog/VLSI:** CMOS Analog Design, Custom Design Flows, Circuit Simulation, SKY130 PDK, Device Matching, DRC/LVS/ERC/PEX, Corner & Monte Carlo Analysis, Analog/Mixed-Signal Design Methodologies
- **Embedded & Digital:** Embedded Systems, IoT, FPGA Architecture, PCB Design, Sensor Interfacing, Finite State Machines (FSM), Testbench Design
- **Programming:** C/C++, Python, Verilog, HTML
- **Software Tools:** Xilinx Vivado, STM32CubeIDE, Proteus, Icarus Verilog, Arduino IDE, LTSpice, GTKWave, Tinkercad

Experience

Analog IC Design Trainee Intern *June 2026*
Sophrosyne Semiconductor, Kunjibettu, Udupi

- Achieved full-cycle understanding of the analog IC design flow by studying schematic design, layout, physical verification, and post-layout validation, resulting in the ability to independently trace a design from concept to tape-out readiness.
- Achieved design-rule compliant layouts by applying DRC, LVS, ERC, and PEX methodologies within industry-standard PDKs and technology/device libraries, resulting in verified, manufacturable CMOS circuit blocks.
- Achieved improved circuit robustness understanding by analyzing common-centroid layout, guard rings, device matching, corner analysis, and Monte Carlo analysis on analog building blocks (MOSFETs, current mirrors, differential amplifiers), resulting in stronger fundamentals in analog/mixed-signal design methodologies.

Projects

- **Bandgap Reference Circuit– Analog IC Project** (ongoing) *SKY130 PDK • CMOS Design • Circuit Simulation*
  Achieved a temperature-stable reference voltage of **1.2V** by designing a CMOS bandgap reference circuit in the SKY130 open-source PDK, resulting in a low-drift voltage source suitable for analog/mixed-signal design applications.
- **Adaptive Real-Time Traffic Light Control – FPGA Project** *Zedboard • FSM • Verilog*
  Achieved zero-defect verified signal integrity by designing FSM-based adaptive traffic control logic in Verilog HDL on a ZedBoard (Zynq-7000 SoC), resulting in a functional real-time signal management system validated via GTKWave simulation.
- **EnviroGuard – IoT Air Quality Monitor** *STM32 • IoT • Proteus*
  Achieved accurate real-time tracking of environmental metrics by integrating STM32 microcontrollers with precision sensors, resulting in a cost-efficient IoT air quality monitoring system validated through structural and budget planning.

Workshops & Certifications

- Embedded System Communication Protocols – TBI GEC Sreekrishnapuram (2026)
- Introduction to IoT and Embedded Systems – University of California, Irvine (Coursera, 2025)
- Interfacing with the Raspberry Pi – University of California, Irvine (Coursera, 2025)

Achievements & Leadership

- Volunteered with IEEE Young Buddies 2.0 Program and actively participated in Robotics Club, IEEE, ISTE, and Mulearn technical societies.